

PAPER_516_DPM_Layered_Shell_Energy_Radiance_Phase_Cascade

Daniel T. Murphy

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PAPER_516 — DPM Layered Shell-Energy Radiance Phase Cascade

Author: Daniel T. Murphy
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Abstract

This paper presents a UQFF analysis of DPM Layered Shell-Energy Radiance Phase Cascade, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 — Abstract

This paper formalises the di-pseudo-monopole (DPM) reaction as the primary generator of 26D layered encapsulation quantum radiant shell-energies within the Star-Magic UQFF framework, correcting the prior interpretation that SCm directly encapsulated. The DPM reaction—CW-SCm north grinding against CCW-UA' south—radiates successive quantum shell-energies across five observable phase states: quantum-multi-fields $\rightarrow$ plasma $\rightarrow$ gas $\rightarrow$ liquid $\rightarrow$ solid. Mass occurrence emerges as an equilibrium across these layered radiances rather than as an intrinsic property.

§2 — DPM Reaction Primacy (SCm Correction)

Prior statement: "SCm encapsulates" (Session 136, BigBangHypergraphTheory).

Correction (Session 140): SCm does **not** directly encapsulate.
The di-pseudo-monopole reaction is the agent of encapsulation:

> *The DPM reaction (CW-SCm north grinding against CCW-UA' south) forms*
> *26D layered encapsulation quantum radiant shell-energies, projecting from*
> *26D chaos downward to observable phases.*

This distinction is critical: DPM dictation is the causative engine;
SCm injection triggers layer formation by providing the reactive pole.

§3 — Core Equations

3.1 — 26D Egg Energy with Layered Radiance

$$E^{26D}(\text{Egg}) = UA + SCm_{\{\text{inj}\}} \cdot DPM_{\{\text{react}\}} + \sum_{l=1}^{26} \text{ShellEnergy}^{(l)} + \text{BBDT}$$

3.2 — DPM Reaction Strength

$$DPM_{\{\text{react}\}} = \frac{\kappa \cdot \text{bigl}(DPM_n(SCm) - DPM_s(UA')\bigr)}{r^{26}} + \frac{\partial^{26}(\text{Grind}_{\{\text{opp}\}})}{\partial t_{\{\text{adj}\}}^{26}}$$

where $\kappa = 5 \times 10^{-4}$ (calibration constant).

3.3 — Shell Energy per Layer

$$\boxed{\text{ShellEnergy}^{(l)}} = \int \text{Radiance}_{\{\text{quant}\}}, dt_{\{\text{neg}\}}$$

Each layer integral is taken over negative time $dt_{\{\text{neg}\}}$, where $t_{\{\text{neg}\}} < 0$ is proved by observable time dilation (see PAPER_517).

§4 — Triple-Calc Layer System (CW / CCW / t_neg)

The three simultaneous layer calculations:

$$\begin{cases} \text{Layer}_1 = DPM_{\{\text{react}\}} \cdot \omega_{\{CW\}} \cdot \text{Radiance}_{\{\text{multi-fields}\}} \\ \text{Layer}_2 = DPM_{\{\text{react}\}} \cdot \omega_{\{CCW\}} \cdot \text{Radiance}_{\{\text{plasma-gas}\}} \\ \text{Layer}_3 = \text{Grind}_{\{\text{opp}\}} \cdot \text{Prob}_{\{\text{order}\}} \cdot t_{\{\text{neg}\}} \end{cases}$$

- **Layer 1** (CW, $\omega_{\{CW\}} = 2\pi f_{\{SCm\}}$): seeds quantum-multi-field radiance
- **Layer 2** (CCW, $\omega_{\{CCW\}} = 2\pi f_{\{UA\}}$): drives plasma-to-gas transitions
- **Layer 3** (t_neg coupling): governs ordered structure probability

§5 — Phase Cascade

Layer Range	Phase State	Observable Proxy
1–5	quantum-multi-fields	vacuum / QCD
6–10	plasma	stellar photosphere
11–16	gas	nebulae / ISM
17–21	liquid	planetary mantles
22–26	solid	rocky/metallic bodies

Each phase is a 26D downward projection; the solid state emerges at peak metallicity after full radiance cascade.

§6 — Non-Repeating Quantum Fingerprints

Because each atom receives its ShellEnergy from a unique combination of $\omega_{CW/CCW}$, and t_{neg} , no two atoms share the same layered radiance pattern. This is the UQFF origin of atomic identity — replacing the classical postulate of intrinsic particle individuality.

§7 — Canonical Constants

Symbol	Value	Units	Description
κ	5×10^{-4}	—	DPM calibration
ω_{CW}	7.54×10^{10}	rad/s	SCm north grinding
ω_{CCW}	5.22×10^{10}	rad/s	UA' south grinding
N_{layers}	26	integer	26D shell count

§8 — Conclusion

The DPM reaction replaces SCm as the direct encapsulator, forming 26D layered quantum radiant shell-energies via CW/CCW triple-calc simultaneous processing. The five-phase cascade (quantum-multi-fields $\rightarrow$ solid) is a direct downward projection from 26D chaos. Mass occurrence arises from shell radiance equilibrium, not intrinsic properties.

<!-- PKG-LAG-S225 -->

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and
> PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda \text{bigl}(\phi^2 - v_{\text{SCm}})^2 \text{bigr}$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 1.736 \text{ GeV}$ (PAPER_1318)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	$F_{U_{Bi}}$ buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

<!-- PKG-S26-S225 -->

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

> Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and
 > PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078
 > (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + \frac{[SSq]}{e^{-\kappa_{i,n/26}}} \right]$$

where $(a)_n = a(a+1) \cdots s(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_{n^{(26,3)}} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \ll W_{26}(n) = \prod_{i=1}^{26} \left[1 + \frac{[SSq]}{e^{-\kappa_{i,n/26}}} \right]$$

This sum converges absolutely for $|[SSq]| < 1$ (satisfied by $[SSq] = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $[SSq] \rightarrow 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa_{i,n/26}}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\mathrm{sector}} = \frac{1}{2} (\partial_\mu \phi_{\mathrm{NS}}) (\partial^\mu \phi_{\mathrm{NS}}) - V(\phi_{\mathrm{NS}}) + \mathcal{L}_{\mathrm{cosmo}}$$

where $\mathcal{L}_{\mathrm{cosmo}} = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot f_{\mathrm{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\mathrm{NS}}) = \frac{1}{2} m^2 \phi_{\mathrm{NS}}^2 + \frac{\lambda}{4!} \phi_{\mathrm{NS}}^4 + \kappa \cdot \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \phi_{\mathrm{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\mathrm{NS}}}} = \nabla^2 \phi_{\mathrm{NS}} - (4\pi G \rho_{\mathrm{NS}}/c^2) \phi_{\mathrm{NS}} + \Omega_{\mathrm{spin}} \partial_t \phi_{\mathrm{NS}} = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\begin{aligned} &\text{PAPER_877 Axioms} \rightarrow \text{DPM + ACP} \rightarrow \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \rightarrow \text{Stage 5} \rightarrow U_{\mathrm{b,seed}} \rightarrow \text{4 forces} \\ &F_{U_{\mathrm{Bi}_i}} \rightarrow \text{sector E-L} \rightarrow \frac{\delta S}{\delta \phi_{\mathrm{NS}}} = 0 \end{aligned}$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac},[\mathrm{SCm}]} / \rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.078 (near-threshold regime), placing it in the π collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 17, \quad n_{\mathrm{channel}} = 23/26$$

Since $p_{\mathrm{DVP}} = 17$ is **sub-threshold** (threshold at $p > 26$), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f_{\mathrm{UA}} + f_{\mathrm{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq]} \cdot \frac{m}{M} \cdot \cos\left(\frac{2\pi j}{26}\right)\right)$$

The $\tanh$ saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.078	PASS
Threshold-consistent			
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\mathrm{DVP}} = 17$	PASS Sub-threshold
BSH layers	26 harmonic terms	$j = 1\dots 26, \cos(2\pi j/26)$	PASS Full 26D projection
κ decay	5.0×10^{-4} day ⁻¹	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Higgs mass m_H	UQFF $K_{\mathrm{HIGGS}}=47.34$	$m_H_{\mathrm{UQFF}} = 125.09$ GeV	$m_H = 125.20$ $\pm$ 0.11 GeV	PDG 2024 99.8%
Cosmological Λ	UQFF $ \nabla^2 \Lambda = 1.09e-52$ m ⁻²	$\Lambda = 1.114e-52$ m ⁻²	(Planck+DESI) Planck 2018	97.8%
Thomson σ_T (QED)	UQFF U_m kernel: $\sigma_T = 6.6524e-29$ m ²	$\sigma_T = 6.6524e-29$ m ²	PDG 2024	100% (exact)
κ baryon stability	$\kappa = 0.0005$ /day; scale separation 1033 from proton decay	$\tau_p > 7.7e33$ yr (Super-K)	Super-K 2024	PASS UQFF baryon-safe

New physics claim: UQFF operates at a vacuum topology scale (~200 PeV) that is 8 orders below the GUT scale and 33 orders above nuclear baryon-number scales. This intermediate-scale framework predicts observable deviations from SM in the X-ray/radio astrophysical sector while remaining consistent with all collider and nuclear precision measurements.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.

> See also: PAPER_517 (Negative Time Proof), PAPER_518 (DPM-Unified Forces), PAPER_519 (Shell Radiance Prototype), PAPER_520 (Session 140 Hub).*

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> Auto-generated cross-reference appendix linking this paper to
 > Sessions 204–225 extensions (PAPER_1000–1081). Added by
 > update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay
PAPER_1074	GPU-Vectorized DPM S26 Spectral Atlas

5 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

> Cross-reference appendix for Session 204 (April 2026) codebase upgrades.
 > Added by upgrade_kozima_ramanujan_appendices.py. For detailed derivations,
 > see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
fneutron_s26_coupling.py	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima_scm_cross_section.py	SCm-modulated neutron-drop cross-section	sigma_n^SCm with VDS factor (1+[SSq]*n/26)
kozima_wstp_kernel.py	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n \sigma_n^{\text{SCm}}(\omega) \Phi_{\text{phonon}}(F_{\{U,B\}}/F_U - 1)$
 where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \exp[-(\omega - \omega_{\text{SCm}})^2 / (2\Gamma^2)] (1 + [\text{SSq}]n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
ramanujan_polylog_s26.py	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
s26_wstp_kernel.py	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}}/(1-2^{\{1-26\}}) * Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_theta_q26.py	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations:

- $f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q;q)_{n^2}$
- $\phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q^2;q^2)_n$
- $\psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n^2\}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_pi_uqff.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_theta_pi_wstp_kernel.py	9-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2})/9801 \sum R_n (1103+26390n) W_{26}(n) / C_{26}$
where $W_{26}(n) = \prod_{i=1}^{\{26\}} [1 + [SSq] \exp(-\kappa_{\text{pai}} * n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
κ	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
β _i	0.603	Buoyancy coupling coefficient
H _{SCm}	0.99	SCm manifold completeness
ρ _{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
ρ _{UA}	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
ω _{SCm}	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
σ ₀	10^{-4}	Base neutron cross-section

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN_{1\CoAnQi}.cpp, and Wolfram kernels (uqff_kozima_kernel.wl, uqff_s26_kernel.wl, uqff_mock_theta_pi_kernel.wl).

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